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A Context-Aware Tourism Recommender System Using a Hybrid Method Combining Deep Learning and Ontology-Based Knowledge

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Abstract

The Santurbán paramo is a sensitive high-mountain ecosystem exposed to pressures from extractive and agricultural activities, as well as increasing tourism. In response, this study presents a context-aware recommendation system designed to support sustainable tourism through the integration of deep neural networks and ontology-based semantic modeling. The proposed system delivers personalized recommendations—such as activities, accommodations, and ecological routes—by processing user preferences, geolocation data, and contextual features, including cost and popularity. The architecture combines a trained TensorFlow Lite model with a domain ontology enriched with GeoSPARQL for geospatial reasoning. All inference operations are conducted locally on Android devices, supported by SQLite for offline data storage, which ensures functionality in connectivity-restricted environments and preserves user privacy. Additionally, the system employs geofencing to trigger real-time environmental notifications when users approach ecologically sensitive zones, promoting responsible behavior and biodiversity awareness. By incorporating structured semantic knowledge with adaptive machine learning, the system enables low-latency, personalized, and conservation-oriented recommendations. This approach contributes to the sustainable management of natural reserves by aligning individual tourism experiences with ecological protection objectives, particularly in remote areas like the Santurbán paramo.

Keywords: context-aware recommender system; tourism recommendation; deep learning; hybrid recommendation method; ontology-based knowledge

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1. Introduction

Tourism in the department of Santander has experienced steady development over the past decades, supported by regional infrastructure interventions, such as the Chicamocha National Park, the Cerro del Santísimo Ecopark, and the modernization of Palonegro International Airport. However, despite these advancements, hotel occupancy rates have not exhibited proportional growth, suggesting the need for strategies that promote a more balanced tourism development model with a broader territorial impact. Within this context, modalities such as

ecotourism and scientific tourism emerge as alternatives oriented toward interaction with natural environments and the appropriation of knowledge, attracting visitor profiles increasingly interested in educational and environmentally responsible experiences [1,2].

The Santurbán Moorland presents favorable conditions for the implementation of these tourism modalities. This high-mountain ecosystem fulfills critical ecological functions, such as supplying water to more than fifteen municipalities, including Bucaramanga and its metropolitan area, and harbors significant biodiversity composed of endemic flora and fauna [3,4]. Landscape features such as lagoons, trails, and characteristic vegetation formations represent natural resources with tourism potential. Nevertheless, the territory faces various socio-environmental pressures, including governance conflicts, extractive activities, and agricultural expansion, which pose risks to its conservation [5,6]. Additionally, the sustainable management of these ecosystems in Colombia is constrained by structural factors such as the undervaluation of biocultural heritage and the limited inclusion of local communities in conservation processes [7].

In light of this scenario, the incorporation of emerging technologies can support the articulation of tourism development and environmental conservation objectives. Industry 4.0 offers a set of tools based on artificial intelligence (AI), machine learning (ML), and the Internet of Things (IoT) that enable the design of adaptive solutions capable of operating in real-time and under limited connectivity conditions [8,9]. These characteristics are particularly relevant in high-mountain ecosystems, such as Santurbán, where technological and logistical constraints require robust and autonomous approaches.

This study introduces a context-aware tourism recommendation system based on the combination of deep learning models with structured knowledge through a tourism-specific domain ontology. The system generates personalized recommendations by considering user preferences, geographic location, and environmental attributes of the surroundings. The technological architecture integrates TensorFlow Lite for on-device inference, GeoSPARQL for spatial queries, and SQLite as a local storage engine, ensuring operability in the absence of connectivity. This functionality is particularly pertinent in protected areas where digital infrastructure is limited. In addition to enhancing the visitor experience, the system incorporates conservation-oriented features, such as geolocation-based alerts, recognition of endemic species, and guidance toward low-impact behaviors.

The integration of artificial intelligence and ontological modeling in this proposal contributes to the development of tourism practices oriented toward sustainability and environmental education. The implementation of the system in the Santurbán Moorland illustrates how technology can support the protection of natural heritage, foster ecological awareness among visitors, and strengthen community participation in responsible tourism strategies [10–12].

2. Theoretical Background

2.1. Recommendation Systems Applied to Tourism

Recommendation systems are essential components within contemporary digital platforms, providing mechanisms for personalizing content and services ranging from electronic products to music, entertainment, and social media. Their capacity to detect behavioral patterns and individual interests has led to their adoption by organizations such as Google, Netflix, Facebook, Twitter, and LinkedIn, with evidence of improvements in commercial indicators such as increased sales and user retention [13,14]. In the tourism domain, these tools support travelers in decision-making processes within information-saturated environments by suggesting destinations, accommodations, routes, and activities based on users' preferences, browsing history, and behavioral patterns [15,16].

Classical recommendation methods include collaborative filtering, which infers a user's interests based on the profiles of similar users; content-based filtering, which suggests items similar to those previously selected; and knowledge-based approaches, which rely on explicit rules and domain structures to generate suggestions [17]. In the tourism sector, these techniques have been applied to generate personalized listings of destinations and activities by drawing on reviews, ratings, demographic characteristics, and browsing records from digital platforms [18].

Although recommendation systems emerged in the early 1990s and have evolved in parallel with advances in artificial intelligence, machine learning, and big data technologies, recent studies indicate that many current models primarily focus on the binary interaction between users and recommended items, without adequately incorporating contextual components [19]. This omission is particularly relevant in tourism, where variables such as location, seasonality, weather conditions, or travel companions significantly influence decision-making processes [20,21].

In response to this limitation, the development of more dynamic and semantically enriched methodologies has been proposed, aiming to integrate contextual, spatial, and temporal attributes to enhance the relevance, diversity, and effectiveness of recommendations in complex domains, such as tourism.

Context-Aware Recommendation Systems (CARS)

Context-aware recommendation systems (CARS) extend conventional models by incorporating spatial, temporal, environmental, or social variables that influence user decisions. This integration is particularly relevant in the tourism domain, where factors such as location, weather conditions, time of day, or visitor density shape the selection of destinations and activities [22].

According to [23], context can be classified based on its mode of acquisition (explicit or implicit), its level of observability (direct or inferred), and its dynamism (static or dynamic). In representational models, context is added as an additional attribute of the system, whereas in interactional models, it conditions the interaction between the user and the recommended items [24].

Recent developments in artificial intelligence have driven the evolution of CARS through architectures, such as recurrent neural networks, transformers, and autoencoders, which enable the modeling of nonlinear relationships between contextual variables and user preferences [25]. Similarly, deep reinforcement learning has been applied to the generation of tourist itineraries, incorporating parameters such as weather and congestion at points of interest to balance visitor flows [26].

Practical examples, such as R2Tour, illustrate the usefulness of integrating real-time contextual data—temperature, precipitation, and distance—to increase the relevance of recommendations [27]. In parallel, semantic approaches based on ontologies and activation models have shown effectiveness in adjusting suggestions according to the user's dynamic profile and environmental conditions [28].

Despite these advances, challenges persist regarding the quality and availability of contextual data, the generalization of models across domains, and the efficient fusion of heterogeneous data sources. Current research focuses on designing methodologies that enhance the adaptability of CARS without compromising interpretability or computational efficiency.

2.2. Ontologies for Tourism Information Management

The heterogeneity and dispersion of data in the tourism sector have led to the adoption of formal mechanisms aimed at structuring knowledge, ensuring interoperability, and

facilitating automated reasoning. In this context, ontologies enable the clear representation of entities, relationships, and attributes within the tourism domain [29].

Initial ontological developments focused on general categories to standardize basic concepts. Notable examples include OnTour, developed by the Digital Enterprise Research Institute (DERI) to model accommodation, location, and gastronomy using OWL-DL; the Mondeca Tourism Ontology, based on the thesaurus of the World Tourism Organization with approximately 1000 entries encoded in OWL [30]; the OTA specification from the OpenTravel Alliance, which employs XML to describe events and activities [31]; and Harmonise, an RDF mediator designed within a European project to integrate accommodations and events [32].

Despite their initial usefulness, these ontologies show limitations in capturing more advanced aspects, such as sustainability, cultural heritage, or community dynamics. In response, more specialized schemes have been proposed. Ref. [33] developed KGDAE, a decoupled autoencoder aware of knowledge graphs that integrates cultural and geographic factors within an ontology comprising 23 types of entities and 37 semantic and spatial relationships, thereby facilitating the interpretation of tourism decisions. Ref. [28] introduced RECESO, an ontology-based model that represents user preferences, context, and points of interest through activation and decay algorithms to promote serendipity in a push-based recommendation framework.

From the perspective of urban heritage, ref. [34] designed CURIOCITY, a three-level ontology grounded in UNESCO definitions to represent tangible, intangible, and natural heritage, implemented in RDF repositories and semantic systems. Ref. [35], in turn, proposed an ontology focused on Indonesian tourist villages, encompassing cultural heritage, natural environments, and infrastructure to enhance interoperability in rural destinations. Additionally, ref. [36], in their review of ontologies for smart cities, highlight the fragmentation of reusable artifacts and recommend more integrated and sustainable models for urban services, including tourism.

In view of these challenges, the present study proposes a recommendation architecture that combines an adaptable geospatial ontology—capable of mediating between different semantic schemes and dynamically updating its knowledge base—with context-aware artificial intelligence methods. This approach aims to provide accurate tourism suggestions tailored to visitor profiles in the Santurbán Moorland.

2.3. Hybrid Recommendation Systems

Given the inherent limitations of single-paradigm recommendation models, hybrid systems have emerged as a robust approach by integrating multiple techniques to enhance accuracy and address challenges, such as data sparsity and the cold-start problem [37]. Their core principle lies in the synergistic combination of different algorithms' strengths to compensate for their weaknesses, resulting in more versatile and effective architectures [38].

Hybridization strategies vary in both complexity and methodological focus. One approach involves the integration of classical techniques with knowledge-based models. For example, the model proposed by [39] combines collaborative filtering (CF) and content-based filtering (CB) with an ontology that represents both the learner and learning objects. In this framework, the ontology supplies the initial knowledge required to profile a new user, enabling the system to generate recommendations even in the absence of a rating history. Similarly, the system described by [40] merges cluster-based collaborative filtering with a semantic rule engine (SWRL) operating on a learning style ontology, thus achieving a high degree of personalization.

The advancement of deep learning has led to the development of more sophisticated hybrid architectures capable of processing heterogeneous data sources using specialized

neural networks. A representative example is the NCTR (Neural network to combine Textual and Rating information) model, which employs a Convolutional Neural Network (CNN) to extract contextual features from textual information while simultaneously learning latent vectors from user ratings. These representations are then integrated through a fusion layer and a Multilayer Perceptron (MLP) to model nonlinear interactions and predict final ratings [38].

An even more advanced architecture is TriDeepRec, which incorporates two distinct deep learning models: a Convolutional Autoencoder (CAERS) for processing content data and a Neural Collaborative Filtering (NCF) component for analyzing user behavior. The outputs of both models are merged through an MLP to generate the final recommendation, thereby leveraging both content and behavioral information [37].

These hybrid approaches have shown particular effectiveness in addressing the cold-start problem. For instance, TriDeepRec's CAERS component is designed to operate solely on content data, enabling it to generate recommendations for new users without requiring prior interactions [37]. Likewise, systems such as the one proposed by [39], which rely on ontologies, are capable of inferring a user's initial preferences based on demographic profiles or declared interests.

In summary, hybrid systems—from the combination of classical techniques with ontologies to deep learning architectures like NCTR and TriDeepRec—offer a flexible and powerful framework. Their ability to integrate content, behavioral data, and semantic knowledge enables the generation of more accurate and resilient recommendations, a methodology directly applicable to the complex and dynamic demands of the tourism sector.

2.4. Industry 4.0 and Its Impact on Tourism

The Fourth Industrial Revolution, or Industry 4.0, encompasses technologies that integrate physical, digital, and biological domains to enable intelligent interaction among cyber-physical systems, communication networks, and automation processes [41]. Its key enablers include artificial intelligence (AI), the Internet of Things (IoT), big data analytics, cloud computing, and simulation [42]. In the tourism sector, these developments have given rise to concepts such as "Tourism 4.0," which applies principles of interoperability and real-time processing to service design, and "smart tourism," focused on the digitalization of destination management and visitor experience [41,43].

Within the tourism value chain, AI and big data facilitate the personalization of offers by identifying behavioral patterns and forecasting demand, thereby enhancing itinerary planning and resource allocation [42]. IoT contributes to this process by enabling the remote monitoring of environmental and operational conditions at tourism facilities, supporting more efficient infrastructure management [44].

The relationship between Industry 4.0 and sustainability is reflected in studies on smart ecotourism, which indicate that the integration of AI and IoT can optimize energy consumption and reduce the environmental impact of tourism activities [44]. However, the adoption of these technologies faces challenges related to implementation costs for small and medium-sized enterprises, digital divides across regions, the need for specialized training, and concerns regarding data privacy and security [42,44].

Within this framework, the present study focuses on the design and implementation of a context-aware recommendation system for the Santurbán Moorland. This system aims to enhance the visitor experience while simultaneously promoting practices aligned with the conservation of the region's biodiversity.

3. Methodology

3.1. Domain Ontology Construction

3.1.1. Scope Definition

The ontology developed for the Santurbán paramo in Colombia integrates the essential elements for sustainable tourism. Its primary function is to recommend activities, services, and routes that align with environmental conservation principles. Several key components have been identified for achieving this goal. Points of interest (POIs) encompass a variety of natural sites, including lagoons and trails, alongside services such as hotels and restaurants, and cultural locations such as churches and viewpoints. These POIs' characteristics include GPS coordinates, operating hours, entry fees, and available services. Additionally, contextual and user information considers geographical location and visitor preferences, including desired activities, such as hiking, photography, and accommodation needs.

This ontology provides a structured framework for managing tourism by leveraging GeoSPARQL components to classify the spatial elements. This framework supports effective spatial queries and contextual analysis [45,46]. Relationships between classes illustrate the connections among POIs, routes, and services, ensuring that users receive meaningful recommendations. Ontology also incorporates conservation guidelines, promoting responsible interactions with natural resources [47].

3.1.2. Ontology Construction

The ontology was developed in Protégé and WebProtégé, both leveraging the Web Ontology Language (OWL), to enable a modular definition of classes, relationships, and restrictions. Initially, a hierarchical structure was established to encompass core categories such as points of interest (POIs)—with subclasses for hotels, viewpoints, natural reserves, safe routes, and restaurants—as well as a Services class, covering offerings like transport and tourist guides, and an Environmental Context class, which captures site-specific accessibility information and environmental constraints [48].

Figure 1 depicts this ontology structure, illustrating the interconnections among critical classes. In particular, the diagram emphasizes how the Tourist Services class links accommodations, camping sites, and safe routes with natural attractions. Moreover, additional classes—including police stations and hiking areas—augment the ontology by representing safety measures and outdoor activities, thereby ensuring comprehensive coverage of the tourism domain.

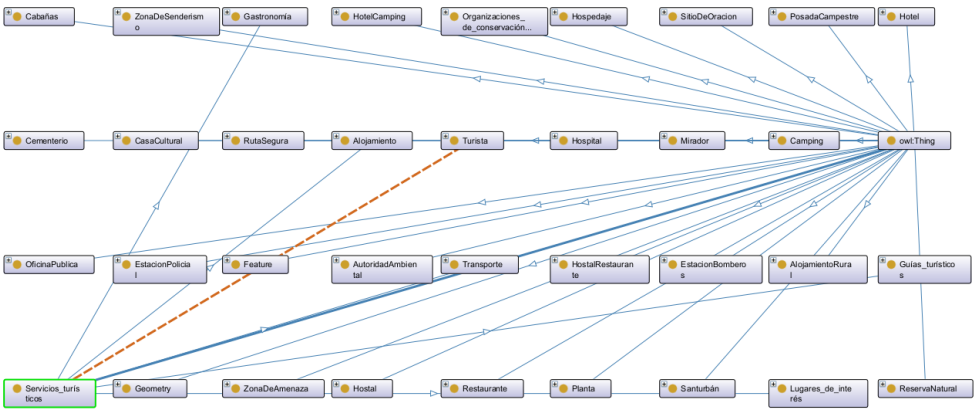

Figure 1. Ontology structure illustrating the relationships between key classes.

Subsequently, the ontology schema was implemented using OWL 2 and extended with GeoSPARQL components to facilitate geospatial reasoning. Main classes under the Servicios_turísticos superclass include traditional tourism services (e.g., Hotel, Hostal,

Camping, Cabañas, Restaurante), community infrastructure (EstaciónPolicía, CasaCultural, OficinaPública), environmental entities (Organizaciones_de_conservación, AutoridadAmbiental), and ecotourism activities (RutaSegura, ZonaDeSenderismo, ReservaNatural). This organization enables unified querying across heterogeneous tourism data sources.

To populate the ontology, open datasets from officially registered tourism providers were integrated and enriched with information obtained from public websites. In addition, field visits validated the resulting knowledge base, creating a methodological triangulation that preserved semantic consistency while accurately reflecting real-world tourism dynamics.

The ontology defines both object and data properties to link POIs to their attributes. For example, the object property geo:hasGeometry associates each POI with its geometric representation in Well-Known Text (WKT) format, while data properties record attributes such as approximate cost, rating, and contact details. Furthermore, class-specific restrictions were imposed to maintain coherence; natural reserves, for instance, are restricted to include hiking areas as their associated subclasses [49].

Finally, by integrating GeoSPARQL, the ontology models' geometries—such as POINT and LINESTRING—that support precise representation of locations and routes. This spatial data not only enhances the system's capability to generate geographically relevant recommendations but also underpins offline navigation functionality, which is essential in remote environments such as the Santurbán paramo [50].

3.1.3. Ontology Population

The ontology was populated using data from multiple sources, including public databases and websites, that provide information on natural reserves, accommodations, and routes. Geolocation APIs were used to obtain precise coordinates for POIs, accommodations, and trails, covering specific sites, such as viewpoints, or natural features, such as lagoons. Environmental reports and tourist guides contributed to conservation guidelines and recommendations for responsible interactions with flora and fauna. The system highlights endemic plant species, identifies their locations along routes to enhance visitor awareness, and encourages responsible tourism practices.

Each POI was populated with detailed instances, which included the name of the site or service, GPS coordinates specified using the WKT syntax, and information on amenities such as Wi-Fi availability, pet policies, and breakfast options. Based on visitor feedback, ratings were collected on a scale of 1–5, while descriptions of safe routes provided information on their duration, high-security zones, and emergency points.

Beyond the elements illustrated in the figure, ontology also includes additional classes relevant to sustainable tourism management. These encompass environmental organizations, conservation practices, and mechanisms for collecting user feedback, forming a comprehensive framework supporting sustainable tourism promotion within the region [48,49].

3.2. Data Generation and Preprocessing

During the learning phase of the semantic network, Python tools, specifically the rdflib library [51], were employed for the extraction and preprocessing of the ontological data [52,53]. This procedure was developed to format the data according to the recommendation system's specific requirements, which are context aware and focuses on the Santurbán paramo in Colombia. As described in the previous section, ontology integrates essential elements for sustainable tourism and plays a fundamental role in providing relevant recommendations based on proximity and user preferences. During data extraction, concepts and relationships from the ontology were transformed into numerical vectors, including key features such as normalized distance (norm_distance), rating (calcification), and normalized cost (norm_costo).

In this system, the normalized distance was assigned the highest weight, as proximity to the tourist's location or the location specified in the search is critical for delivering precise recommendations. The second most important factor was rating, which reflects the place's popularity based on the number of reviews, average rating, and frequency of visits. Finally, cost (norm_costo) was the factor with the least weight among the three but remained relevant for users seeking options within specific budget ranges.

Preprocessing involved imputing missing values using the SimpleImputer class and normalizing the variables using StandardScaler or MinMaxScaler. These scalers were saved with Joblib for reuse, ensuring data consistency throughout the inference process. The data were divided into training and validation subsets, allocating 80% and 20% of the data for training and validation, respectively [54]. This division guarantees that the dataset includes a representative segment for model training, preserving an adequate portion to assess performance on previously unseen data. The code implementation employed the train_test_split function with a test_size parameter set to 0.2 and a random_state value of 42 to guarantee reproducibility [55].

Both datasets were converted into TensorFlow datasets to streamline the training process. The training dataset was shuffled with a buffer size of 1024 to avoid order bias and was divided into batches of 32 samples each, ensuring efficient processing during model training. Similarly, the validation dataset was prepared in batches of the same size to maintain consistency across the evaluation stages. This approach optimizes resource usage and ensures that the model generalizes effectively, thereby avoiding overfitting [56].

The model architecture was implemented using Keras as a Deep Neural Network (DNN) designed to capture the semantic relationships between concepts in the ontology. The network consists of multiple dense layers with nonlinear activation functions, enabling the modeling of complex relationships within the data. The first layer contained 128 neurons with a ReLU activation function, followed by a dropout layer with a rate of 30% to prevent overfitting [57]. The second layer consists of 64 neurons activated using the ReLU function, followed by an additional dropout layer configured with an identical rate. The output layer consists of a single neuron with a linear activation function that is suitable for regression tasks [58,59].

Initially, the model was compiled using the Mean Squared Error (MSE) loss function and optimized with the Adam algorithm at a learning rate of 0.001. Training was scheduled for 300 epochs, incorporating an EarlyStopping callback that monitors validation loss and halts the process if no improvement occurs after 25 consecutive epochs, subsequently restoring the best-performing weights.

Subsequently, model performance was assessed through a suite of regression metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), and the coefficient of determination (R^2). These measures were chosen to evaluate both the accuracy and the robustness of predictions across varying value ranges and user scenarios.

Finally, the trained model, together with its associated scalers and imputers, was exported using Joblib and the TensorFlow SavedModel format. This configuration ensures compatibility with TensorFlow Lite for mobile inference, thereby facilitating real-time, offline recommendations on Android devices.

3.3. Development of the Recommendation System

The Android recommendation system is based on a mixed model that integrates deep neural networks with a hybrid algorithm that combines a semantic network and collaborative filtering techniques [57]. The neural network generates recommendations aligned with user preferences, which are stored locally on the device by using SQLite [60].

This approach ensures data privacy because all personal information remains on the device without requiring synchronization with external servers. The system is designed to operate online and offline, enhancing usability in remote areas, such as the Santurbán paramo.

3.3.1. System Overview

The system architecture, illustrated in Figure 2, integrates an interactive user interface with a modular set of components that operate both online and offline. These components include a semantic inference engine, a hybrid recommendation algorithm, a local database for user preferences, and spatial reasoning capabilities based on ontological structures.

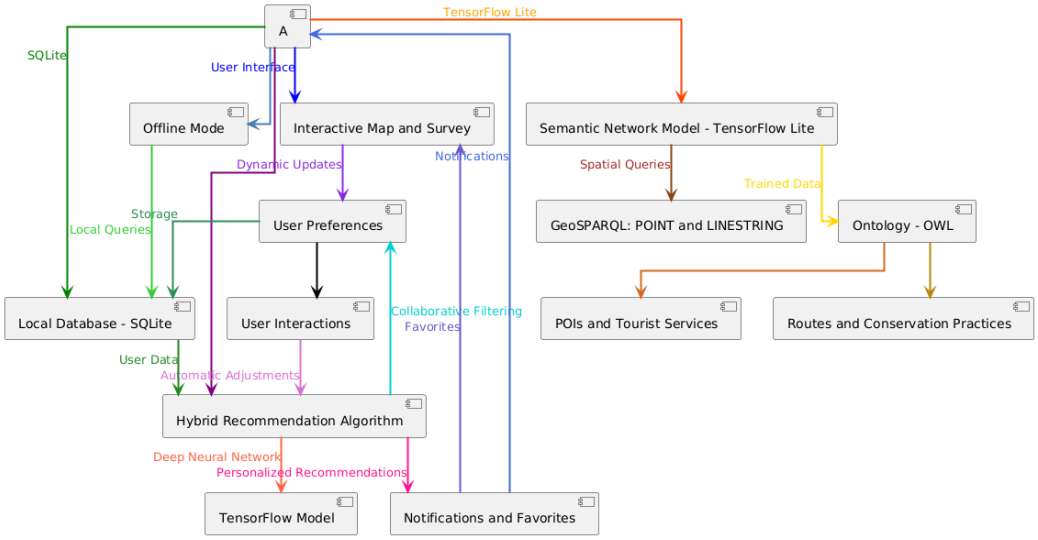

Figure 2. Architecture of the recommendation system.

The general workflow begins with user interactions—such as filling out surveys or navigating the POI map—which are recorded locally and used to adjust the user's preference profile. These data are then processed by a hybrid algorithm that combines the outputs of a deep neural network with semantic reasoning and collaborative filtering. The system delivers personalized recommendations based on a combination of objective factors (e.g., proximity, popularity, cost) and subjective aspects (e.g., preferences, behavioral history).

Spatial analysis is supported by GeoSPARQL-enhanced ontologies, enabling the application to identify and prioritize POIs within relevant geographic zones. Results are presented via a responsive and user-friendly interface that enables route planning, bookmarking of favorites, and interaction with nearby POIs.

3.3.2. Integration of the Semantic Network

The semantic network is trained with data extracted from an ontology that represents relationships between points of interest (POIs), tourism services, ecological routes, and conservation principles. This ontological foundation supports contextual analysis through GeoSPARQL queries, ensuring that recommendations are aligned with both geographic location and user intent.

Geometries defined using POINT and LINESTRING are embedded in the ontology to model both POI locations and route structures. These are parsed and evaluated to filter recommendations according to spatial relevance. The semantic model is deployed on mobile devices using TensorFlow Lite and is accompanied by locally stored scalers and imputers to support real-time inference without external dependencies—an essential feature in regions with limited connectivity, such as the Santurbán paramo [61].

3.3.3. Hybrid Recommendation Algorithm

The recommendation system combines the neural network output with the user preferences stored on the device, following a hybrid approach [62]. User preferences are recorded in a local database and dynamically updated based on interactions. The corresponding category's score increases if the user engages with a recommendation after at least four hours. Conversely, if no interaction occurs with a category within one day, its score decreases, allowing the system to adapt to the evolving user interests [63].

The neural network determines the initial score by prioritizing proximity as the primary factor, followed by site popularity (ratings), and cost as complementary criteria. This initial ranking is then combined with the stored preferences of the user [64,65].

The final score for each site was calculated by merging the predicted values from the semantic-network-based model with the user's preferences. A weight of 60% was assigned to the model's predictions, whereas the remaining 40% reflected the user's personal preferences. This weighting ensures that the recommendations account for objective model analysis and individual user interests, prioritizing sites that align with both criteria.

User Interface [66].

The user interface developed in Flutter offers an intuitive way of interacting with the recommendation system. Users can visualize POIs on a map and plan online and offline routes, ensuring functionality in areas with limited or no connectivity. An initial survey module allows users to define their preferences, and a setting section enables modifications at any time, ensuring that the recommendations remain aligned with their needs.

The application records each user's interaction with the recommendations and automatically adjusts future suggestions based on the updated preferences. Users can also mark sites as favorites and receive notifications about relevant nearby locations [16].

3.3.4. Data Management and Offline Operation

SQLite is a local database that manages preferences and interactions without requiring Internet access to ensure data privacy. All the information is stored exclusively on the device, eliminating the risks associated with external server usage. This dynamic management allows the system to adjust real-time recommendations as user interests evolve. The overall structure of the system is presented in Figure 2.

The recommendation system combines deep neural networks with collaborative and content-filtering techniques to provide contextual and personalized recommendations [67,68]. Operating offline ensures functionality in remote areas, such as the Santurbán paramo, while integrating user preferences in a local database safeguards privacy. This approach balances personalization, performance, and sustainability by promoting regional tourism.

3.3.5. System Implementation and Data Flow Integration

To ensure robust operation in both connected and disconnected environments, the system integrates three core components—namely, a local database (SQLite), a geospatial reasoning module (GeoSPARQL), and a trained semantic model deployed via TensorFlow Lite—within a coordinated pipeline orchestrated by the user interface and the hybrid recommendation algorithm.

In offline mode, when connectivity is unavailable—as is common in remote areas such as the Santurbán páramo—the application executes all logic locally. First, user interactions (e.g., marking favorites or dismissing recommendations) and survey responses are recorded in SQLite. Next, these inputs adjust the user profile, updating the scores assigned to various POI categories. Simultaneously, the hybrid algorithm queries the pre-trained TensorFlow Lite model, using normalized features (distance, rating, cost) together with the current profile, thereby generating a ranked list of recommended sites.

Spatial reasoning is enabled through custom Python utilities embedded in the app, which parse the WKT geometries from the ontology and simulate GeoSPARQL queries. This allows the system to perform geospatial filtering by evaluating whether POIs intersect user-defined zones or lie within specified proximity thresholds—without requiring an external SPARQL endpoint.

In online mode, the same core operations are executed, with the addition of dynamic enhancements—such as map tile synchronization and real-time alerts—delivered via internet services. Finally, the data flow follows a bidirectional sequence: from user interactions to preference adjustments in SQLite, onward to inference via TensorFlow Lite, then to geospatial validation through the ontology-based spatial filter, and ultimately to rendering the results on the interactive map. Figure 2 illustrates this architecture and the sequence of interactions.

3.4. Mathematical Summary of the Recommendation System

3.4.1. Contextual Data Extraction and Geo-Distance Calculation

Let I be the set of POIs (Points of Interest) considered by the recommendation system, which are extracted from the ontology of tourism. Each point of interest belonging to i I is represented in the RDF knowledge base by data properties that include its geographic location (stored as WKT geometry), an average rating provided by users, and an estimated cost value.

The geographical location of each POI is expressed by its latitude and longitude coordinates. Formally, for a point of interest i, its location is defined as a tuple:

$$L(i) = (\varphi_i, \lambda_i)$$

where φ_i and λ_i denote the latitude and longitude of the POI, respectively, both expressed in radians.

Similarly, the location of user u is also represented by its latitude and longitude, i.e.:

$$L(u) = (\varphi_u, \lambda_u)$$

where φ_u and λ_u represent the latitude and longitude of the user, respectively.

With this information, between the user and each POI. This distance is a key element of the proximity context used in the recommendation model. To calculate this distance between the user and the point of interest, Haversine's formula is used. This equation enables calculating the distance between two geographical points on the surface of a sphere based on their coordinates: the system computes the geographical distance.

$$d(u,i) = 2R.arcsin\left(\sqrt{sin^2\left(\frac{\varphi_i - \varphi_u}{2}\right) + cos(\varphi_u)cos(\varphi_i)sin^2\left(\frac{\lambda_i - \lambda_u}{2}\right)}\right)$$

where represents the mean radius of the Earth. This distance value is a direct measure of the proximity between the user and the point of interest and is incorporated as a key contextual factor in the recommendation system $R \approx 6371$ km.

3.4.2. Attribute Normalization and Imputation

Once the raw data for each point of interest (POI) has been obtained—such as distance to the user, the average rating given by other users, and the estimated cost—a preprocessing process is performed to fill in the missing values and normalize the numerical attributes.

Suppose a POI $i \in I$ lacks a rating r_i or an estimated cost c_i. In these cases, an imputation function is used that employs the average of the $\mathcal{S}$ to estimate the missing value. Formally, the imputed qualification r_i is defined as:

$$r_i \leftarrow \begin{cases} r_i, & \text{if } r_i \text{ is present} \\ \frac{1}{|I|}\sum_{j\in I} r_j, & \text{If it's unknown} \end{cases}$$

Similarly, if the value of is not available, it is allocated using the average of available costs c_i.

$$c_i \leftarrow \begin{cases} c_i, & \text{if } c_i \text{ is present} \\ \frac{1}{|I|}\sum_{j\in I} c_j, & \text{If it's unknown} \end{cases}$$

After imputation, the numerical variables are normalized to bring them to a uniform scale, thus avoiding any individual variable disproportionately influencing the recommendation outcomes. For this purpose, Min–Max standardization is used. For example, cost normalization c_i is expressed as:

$$\widetilde{c_i} = \frac{c_i - \min_{j\in I} c_j}{\max_{j\in I} c_j - \min_{j\in I} c_j}$$

This same procedure is applied to normalize the geographical distance di and the grade r_i, obtaining di and r_i, respectively. This normalization ensures that all attributes are scaled to a common range, typically between 0 and 1, which is essential to avoid biases caused by scale differences.

Finally, each point of interest i is represented by a vector of normalized characteristics of the form:

$$x_i = \left(\widetilde{d}_i, \widetilde{r}_i, \widetilde{c}_i\right)$$

This vector is the final input to the recommendation model, which will use this representation to calculate the relevance of each POI according to the user's context. which uses it to estimate the relevance of each POI with respect to the user's context.

3.4.3. Deep Neural Network Inference

To estimate a recommendation score for each point of interest (POI), a dense neural network is trained. It takes as input the POI feature vector, represented by x_i This vector contains normalized information related to the distance to the user, the average rating, and the estimated cost of the POI.

The neural network implements a mathematical function of the type:

$$f : R^3 \rightarrow R$$

Indicating that the network processes a three-dimensional input vector and produces a scalar output, representing the recommendation score for the POI.

During the training stage, this feature f is tuned to minimize a prediction error on a training dataset. This process allows learning a set of internal parameters of the network, commonly referred to as weights and biases, which are grouped into a parameter vector θ.

Once the neural network is trained, it is used in the inference phase, where parameters are frozen (i.e., no longer modified) and represented by θ^*. At this stage, the model predicts the recommendation score of a POI by applying the learned function on its feature vector:

$$s_i^{\mathrm{NN}} = f(x_i; \theta^*)$$

where

- ${s_i}^{\mathrm{NN}} \in \mathbb{R}$ is the scalar value that represents the estimated utility of the point of interest, as predicted by the model.
- $x_i \in \mathbb{R}^3$ is the input vector that contains POI information.
- θ^* It represents the weights and biases of the network after training.

This value ${s_i}^{\mathrm{NN}}$ is finally used by the recommendation system to sort or filter the points of interest according to their relevance to the user.

3.4.4. Integration of User Preferences and Generation of the Final Ranking

Each user has a local preference profile, which is modeled by a function that assigns a preference value for each category of points of interest. This function is denoted as: (POIs).

$$ {s_i}^{\mathrm{NN}} = f(x_i; \theta^*) $$

where c represents a specific category of POIs such as hiking, accommodation, and photography, among others. The value u expresses how much interest the user has in the category c, with the minimum interest 0 being the maximum interest 1.

For each poi i, if it denotes $cat(i)$ its primary category, then the user's preference score towards that one poi is defined as:

$$ {s^{(\mathrm{pref})}}_i = p_{\mathrm{u}}, cat(i) $$

This value reflects the user's affinity with the type of experience offered by them POI.

The system then combines the score predicted ${s^{(\mathrm{NN})}}_i$ by the neural network with the user's preference ${s^{(\mathrm{pref})}}_i$ using a weighted formula that balances both factors:

$$ {s^{(\mathrm{final})}}_i = \alpha \cdot {s^{(\mathrm{NN})}}_i + (1 - \alpha) \cdot {s^{(\mathrm{pref})}}_i $$

where the parameter $\alpha \in [0, 1]$ controls the relative importance of the model versus personal preferences. In practice, the value of the following is typically used:

$$ \alpha = 0.6 $$

This configuration assigns greater importance to the model prediction while still accounting for the user's stated preferences.

Finally, a personalized ranking ${s^{(\mathrm{final})}}_i$ is obtained by sorting items in descending order based on their final scores. That is, the list of $L(u)$ recommendations is defined as:

$$ L(u) = \left[i^{(1)}, i^{(2)}, \ldots, i^{(\mathrm{N})} \right] \textit{such that } {s^{(\mathrm{final})}}_{i^{(1)}} \geq {s^{(\mathrm{final})}}_{i^{(2)}} \geq \ldots \geq {s^{(\mathrm{final})}}_{i^{(N)}} $$

This list represents the N personalized recommendations for the user, incorporating both contextual information (from the neural model) and explicit user preferences.

4. Results

4.1. Evaluation of the Deep Learning Model

A set of metrics encompassing accuracy, relevance, and generalization capability was implemented to assess the recommendations generated by the TensorFlow deep learning model using RDF data from the ontology. This evaluation considered both regression-based and ranking-based indicators to examine not only the numerical consistency of predictions but also their order of relevance in practical use cases [69–71].

4.1.1. Mean Squared Error (MSE)

Minimizing significant prediction deviations is crucial in machine-learning-based recommendation systems. The Mean Squared Error (MSE) penalizes large-magnitude errors, ensuring that recommendations, such as estimated distance or cost, do not exhibit excessive deviations.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where y_i is the actual value, $\hat{y}_i$ is the predicted value, and n is the total number of observations.

4.1.2. Mean Absolute Error (MAE)

The Mean Absolute Error (MAE) assesses the model's average accuracy, minimizing the disproportionate influence of outliers. In the context of this project, MAE is useful for determining the absolute distance between the generated recommendations and user expectations.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

4.1.3. Root Mean Squared Error (RMSE)

In the TensorFlow model for a context-aware recommendation system that considers distance, rating, and cost, the Root Mean Squared Error (RMSE) is particularly relevant. Representing the error in the same units as the predictions simplifies the interpretation of its scale. In this model, which recommends points of interest (POIs) based on GPS coordinates, ratings, and costs, the RMSE allows the evaluation of the overall accuracy of the recommendations.

$$RMSE = \frac{1}{n}\sum_{i=1}^{n}\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

4.1.4. Coefficient of Determination (R^2)

The coefficient of determination $(\boldsymbol{R}^2)$ measures the proportion of variability in the data that the model explains. In the context of a recommendation system, a high $(\boldsymbol{R}^2)$ value suggests that features such as location, cost, and ratings are relevant predictors of the generated recommendations.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \overline{y})^2}$$

where $\overline{y}$ is the mean of the actual values.

4.1.5. Mean Absolute Percentage Error (MAPE)

Given that this project involves data of different scales, such as cost ranges or distances, the Mean Absolute Percentage Error (MAPE) is a valuable metric for comparing the model's performance in relative terms. By expressing the error as a percentage, the MAPE facilitates a more equitable comparison between predictions involving different units of measurement.

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

4.1.6. Area Under the ROC Curve (AUC)

The Area Under the ROC Curve (AUC) is a fundamental metric in recommendation systems that classify items according to their relevance or likelihood of user interest. In this context, AUC evaluates the model's ability to discriminate between relevant and non-relevant points of interest (POIs).

The AUC is interpreted as the probability that the model assigns a higher relevance score to a POI that should be recommended compared to one that should not.

Additionally, ranking metrics were employed to evaluate the ability of the model to order predictions based on their relevance. The ranking metrics used included the following:

- Precision@k: Represents the proportion of relevant items among the top k recommended items.
- Recall@k: Indicates the fraction of relevant items retrieved from the top k recommended items.
- NDCG@k: measures the normalized cumulative gain based on relevance, considering the position of items in the ranking. This metric offers a more comprehensive evaluation by penalizing relevant recommendations in lower-ranking positions.

The selection of this set of metrics ensures that the recommendation system is not limited solely to minimizing errors, but also provides relevant predictions aligned with user expectations. Considering that the system operates offline, the accuracy of recommendations from the first interaction becomes crucial, as there is no possibility of online corrections.

4.2. Model Results

4.2.1. Training Conditions and Performance Summary

The training process was conducted in a local environment with an Intel® Core™ i5-6400 processor (2.70 GHz), 16 GB RAM, and Windows 10 Pro (version 2H2). The model was implemented using Python 3.7, with TensorFlow and Keras handling the deep learning workflow, and Scikit-learn used for preprocessing tasks.

The training spanned 50 epochs, using the Adam optimizer with a learning rate of 0.001 and an Early Stopping mechanism to monitor validation loss. Upon completion, the model reached a final validation MSE of 0.0039 and MAE of 0.0508, indicating convergence without overfitting [71].

4.2.2. Regression Outcomes and Residual Behavior

Quantitative evaluation on the validation subset produced the following outcomes: RMSE = 0.1955, MAE = 0.0508, MSE = 0.0039, and R^2 = 0.9959. The MAPE metric further supported performance consistency across varying scales [69,70]. These results reflect that the trained model maintained stable predictive behavior when estimating the relevance of points of interest (POIs) based on geospatial and semantic features. Given the system's offline nature and its intended deployment in connectivity-restricted environments such as the Santurbán paramo, these levels of accuracy are relevant because they reduce the likelihood of misguiding users in critical decision moments, such as selecting routes or services. The low RMSE and MAE suggest that predicted scores remain close to actual values, while the high R^2 confirms that contextual features like location, rating, and cost contribute significantly to explaining recommendation relevance within the model's structure.

The residuals, shown in Figure 3, appear evenly distributed, with no visible heteroscedasticity. This suggests that prediction errors are stable across the range of output values. The ROC curve depicted in Figure 4 yielded an AUC of 0.99, which supports the model's ability to differentiate between relevant and non-relevant items based on the input features.

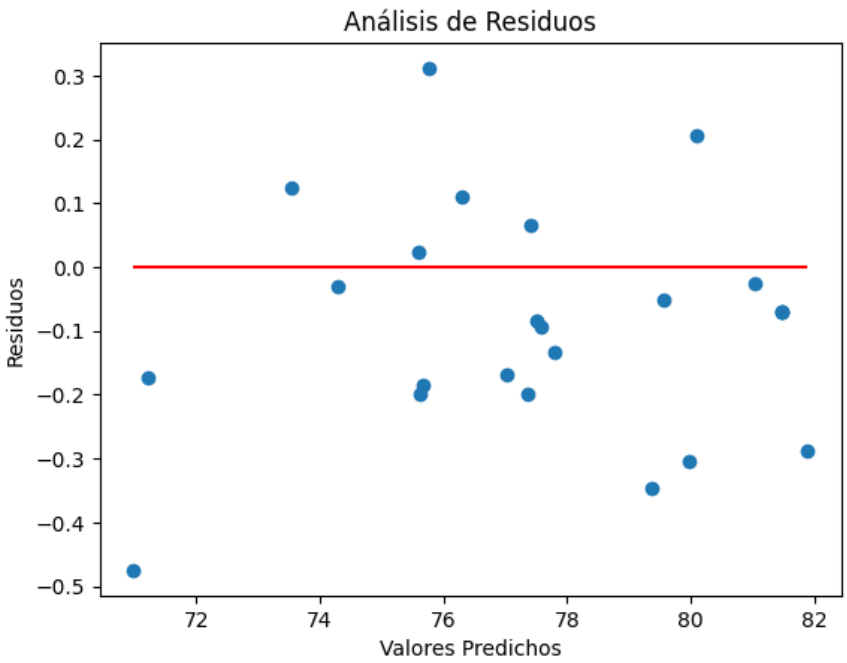

Figure 3. Residual analysis.

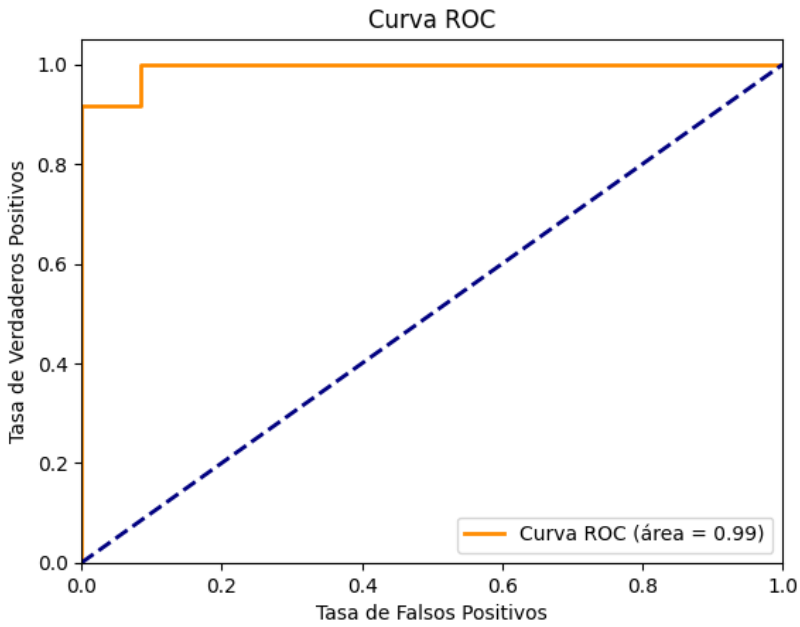

Figure 4. ROC curve.

4.2.3. Model Conversion and Mobile Deployment

Following the training and validation stages, the final deep learning model was exported in h5 format and subsequently converted to the tflite format using the TensorFlow Lite converter. This conversion enables model inference directly on Android mobile devices, allowing the recommendation system to function independently of cloud infrastructure and internet connectivity [72–74]. The TensorFlow Lite model is deployed alongside previously saved normalization scalers and imputers, enabling seamless integration into the mobile application's inference pipeline.

Once deployed, the model operates as part of a hybrid algorithm that processes contextual input—including geolocation coordinates, user preferences, and predefined spatial filters—to produce relevance scores for candidate POIs. The app uses this information to update rankings based on both predictive and preference-based criteria.

4.3. Implementation and Results Site Recommendations in Santurbán

This study presents a context-aware recommendation system developed as a technological tool for tourists visiting the Santurbán Páramo region. The system aims to promote environmental conservation by generating tourism recommendations that encourage sustainable and responsible use of the natural environment. The implementation focused on the municipalities of California, Vetas, Suratá, Tona, and Matanza, including points of interest (POIs) in Bucaramanga and other representative sites in the Santander department. Figure 5 illustrates the geographical location of the POIs integrated into the ontology.

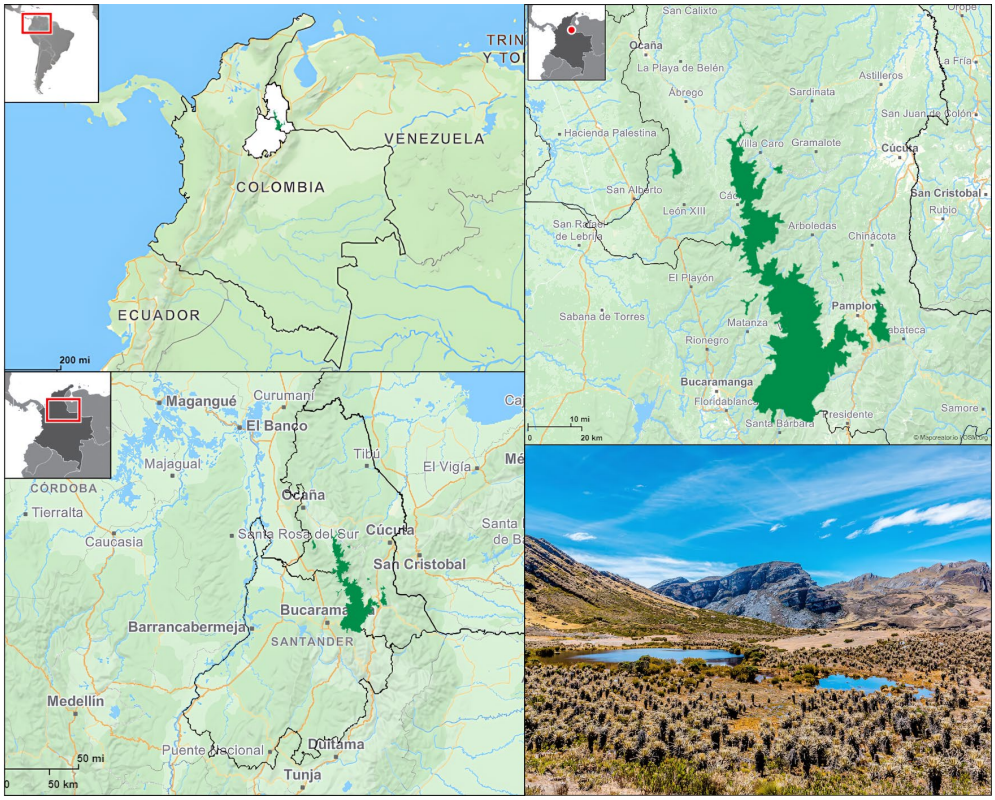

Figure 5. Geographical location of the system's points of interest.

POIs are semantically structured in the ontology into categories such as accommodation, gastronomy, safe routes, and natural attractions [1,75]. Enabling both contextual filtering and classification by relevance. User profiles are initialized through a survey interface in which users assign numerical values (0–100) to preferences associated with specific tourism activities and services (e.g., hiking, observing endemic flora, staying in cabins, or trying local dishes) [76]. These values are stored in the local database and are dynamically updated as users interact with the application.

The system operates through a multi-stage process, shown in Figure 6, which outlines the data flow from user input to recommendation output. The process includes: (1) Data Collection, where user preferences and GPS coordinates are gathered; (2) Feature Vector Preparation, in which contextual filters and normalized attributes are applied (distance, rating, cost); (3) Model Inference, where the TensorFlow Lite model estimates relevance scores for each POI; (4) Preference Fusion, where these scores are weighted against the user's preference profile; and (5) Output Visualization, which presents ranked recommendations to the user on a map-based interface.

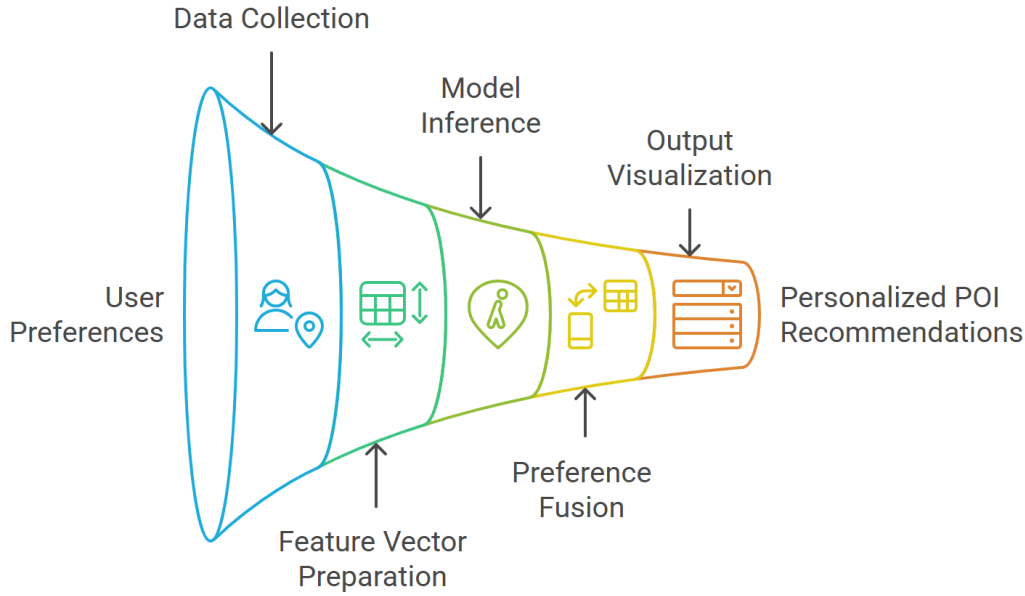

Figure 6. Tourist recommendation pipeline: from user input to ranked results.

Upon authorization, the system uses the device's GPS to determine the tourist's location and feeds this information into the model to evaluate proximity and generate relevance scores. The recommendation algorithm integrates three main contextual dimensions: (1) the user's geographic position, essential in high-altitude terrain such as the paramo; (2) popularity metrics derived from previous user feedback; and (3) estimated cost, when available [68,77].

In the case study presented, a tourist located near the town of Berlín received recommendations by activating the app and allowing location access. The system identified nearby POIs using their coordinates, computed their feature vectors, and generated a preliminary ranking based on contextual relevance. The final score for each POI was computed by weighting 60% of the prediction obtained from the neural network and 40% from the user's stored preferences [78].

For this specific user, higher weights were assigned to hiking trails, natural reserves, and viewpoints. Camping and cabin lodging received intermediate weights, while institutional facilities were deprioritized. This prioritization was reflected in the final ranking presented in Table 1, where routes and natural sites achieved the top positions. For example, the "Sendero a la Laguna Pajarito" route scored 74.03, and "Santurbán Paramo," a natural reserve, scored 75.17—both aligning with the user's stated preferences.

The recommendation workflow, reinforced by spatial and semantic inference, allows for high alignment between suggested sites and individual user expectations. This combination of predictive and personalized components enhances the contextual relevance of suggestions, especially in remote zones with limited connectivity. This functionality is illustrated in Figure 7, which presents the location of the recommended points centered on the user's position.

Table 1. Personalized ranking of points of interest in the Santurbán paramo based on location, popularity, and user preferences.

Recommendation	Site	Category Ranking	TensorFlow Rating	Score Initial	Distance	Final Score
1	Santurbán Moor	Nature reserve	9	79.97	7.67	75.1718
2	Pajarito Lagoon Trail	Route	19	78.75	10.21	74.025
3	Cunta Lagoons Trail	Route	20	78.74	10.22	74.0156
4	El Gramal Viewpoint	Viewpoint	10	79.8	8.04	73.416

Table 1. *Cont.*

Recommendation	Site	Category Ranking	TensorFlow Rating	Score Initial	Distance	Final Score
5	Piedra Parada Refuge	Camping	8	80.29	5.33	70.6552
6	Frailejones Campestre	Cabins	15	79.06	6.43	63.248
7	Where Javis Zairotales	Restaurant	1	82.82	0.11	62.9432
8	Amparito Market	Restaurant	2	82.52	0.66	62.7152
9	El Esquinazo Berlinero	Restaurant	3	82.44	0.08	62.6544
10	La Berlinera	Restaurant	4	82.28	0.46	62.5328
11	Disterigma Empetrifolium	Plant	17	78.83	9.9	61.4874
12	Senecio Niveoaureus	Plant	18	78.82	9.91	61.4796
13	The Delights of Diana	Restaurant	11	79.46	9.78	60.3896
14	El Fogón De Los Ramírez	Restaurant	12	79.38	9.86	60.3288
15	El Molino	Restaurant	14	79.3	9.98	60.268
16	Tona Gourmet	Restaurant	16	78.84	9.81	59.9184
17	San Isidro Hospital	Institutional	5	82.18	0.04	52.5952
18	Berlin Police Station	Institutional	6	82.16	0.09	52.5824
19	Berlin Cemetery	Institutional	7	82.02	0.51	52.4928
20	Checkpoint1	Institutional	13	79.36	7.67	50.7904

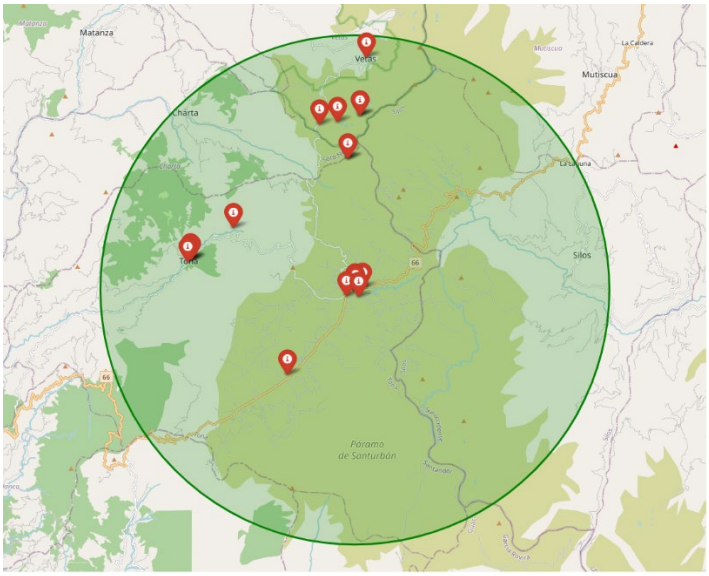

Figure 7. Location of the recommended points centered on the user's location.

4.3.1. Recommendations for Environmental Sustainability

As part of its support for responsible tourism, the recommendation system integrates background GPS tracking and ecological awareness mechanisms to provide conservation-oriented guidance within the Santurbán Paramo. With user authorization, the application continuously monitors location data to enable real-time context-aware interventions, even when operating in the background or with the screen locked.

To minimize ecological disturbance, the system employs geofencing techniques that define virtual boundaries around environmentally sensitive areas, such as habitats with endemic flora, high-altitude wetlands, and protected ecological zones. When users enter these predefined perimeters, the application generates notifications promoting sustainable behaviors, including adherence to marked trails, noise reduction, and respectful interaction with local biodiversity [5].

Beyond alerts, the system also delivers site-specific conservation messages intended to raise user awareness. For instance, upon nearing locations such as black lagoons, users receive contextual information about local vegetation, including characteristics of native frailejón species, and are reminded of the importance of ecosystem preservation [79]. These recommendations are presented as location-triggered notifications, designed to inform and influence visitor behavior through timely and relevant content.

Figure 8 illustrates the designated safe route toward the Black Lagoons, highlighting the ecological checkpoints embedded within the application. The corresponding conservation messages and geolocated markers are detailed in Table 2, which includes both biological insights and behavioral guidance for minimizing visitor impact.

Figure 8. Safe route to the Black Lagoons.

Table 2. Marked points and conservation recommendations on the Safe Route to Lagunas Negras.

Point	GPS Coordinates	Type of Recommendation	Message
1	7.283326–72.894961	Conservation	Dear visitor, upon entering this protected area, please stay on authorized trails, avoid harming the flora and fauna, and take all waste with you. Your respect and care help preserve the beauty and balance of this valuable ecosystem. Thank you for contributing to its protection!
2	7.281290–72.893714	Endemic Plants	You are approaching a medium-sized frailejón (Espeletia), which can grow up to 2 m in height. It is characterized by a stem and a rosette 50 cm in diameter with broad leaves in green and yellow hues. From the center of the rosette, floral stems with yellow flowers emerge. This species is native to Colombia and part of the unique páramo ecosystem.
3	7.279387–72.891833	Conservation	Reminder: you are in a unique and fragile ecosystem. Please do not litter and respect the flora and fauna. Your commitment helps preserve the beauty and balance of this natural environment. Thank you for being a responsible tourist!
4	7.277666–72.889650	Endemic Plants	You are near a frailejón (Espeletia) that can reach up to 60 cm. It has thick, elongated leaves up to 45 cm, narrowing into a pseudo-petiole at the base. Its flowers, in shades of yellow to brown, form an oval-shaped inflorescence. Appreciate its beauty and respect its surroundings to protect this unique species.
5	7.275572–72.885983	Conservation	You have arrived at the Black Lagoons, a natural environment of high ecological value. Please remember that entering the water is not allowed to protect this fragile ecosystem. Enjoy the landscape responsibly and help preserve its beauty for future generations.

The execution of background tasks ensures the continuous delivery of conservation-oriented recommendations without requiring direct user interaction. This feature is particularly relevant in remote or high-altitude areas, where persistent functionality is essential for maintaining communication with the user. Upon entering a geofenced zone, the system emits a non-intrusive auditory message, ensuring that users receive ecological guidance without visual disruption or excessive repetition. This mechanism is designed to minimize discomfort while reinforcing environmentally responsible behavior.

In addition, the system integrates educational content regarding the biodiversity and ecological significance of the Santurbán Paramo. As users approach designated conservation areas, they are informed—through contextual audio cues—about key elements such as endemic flora, hydrological importance, and conservation regulations. This approach supports environmentally conscious decision-making and promotes a deeper understanding of the region's ecological value [80].

4.3.2. User Evaluation at Laguna Negra

To complement the system's technical validation and its ecological recommendation functionality, a field evaluation was conducted along the trail to Laguna Negra, one of the key eco-tourism destinations in the Santurbán Paramo. In this scenario, participants interacted with the application under real environmental conditions and were later invited to complete a brief survey. The evaluation focused on three dimensions: (i) the perceived usefulness of the recommendations, (ii) overall satisfaction with the tool, and (iii) the diversity of the suggestions received. In addition to quantitative responses, open-ended feedback was collected to capture user perceptions regarding usability and content coverage. Table 3 presents a sample of the user evaluations gathered during this test session, illustrating both the numerical ratings and representative user comments.

Table 3. Sample of user evaluations after interacting with the recommendation system at Laguna Negra.

User	Usefulness (1–5)	Satisfaction (1–5)	Diversity (1–5)	Comment
User_1	5	3	5	Very good experience, useful and accurate suggestions.
User_2	4	3	4	I would like more information about flora and fauna.
User_3	4	4	5	Excellent tool for planning the visit.
User_4	4	4	3	It would be helpful to include alternative routes.
User_5	4	4	4	The app worked well even without a signal.
User_6	4	5	3	Good guide, but the interface could be improved.

Overall, participants rated the system favorably across all evaluated aspects. Usefulness achieved an average score of 4.4 out of 5, indicating that recommendations were timely and appropriate for the route context. Satisfaction reached 4.1, and diversity was rated at 4.2, reflecting a positive perception of the variety and contextual relevance of the content. Qualitative feedback emphasized the utility of the system for route planning and real-time orientation, while also suggesting improvements related to the interface and the inclusion of more detailed biodiversity information.

These findings support the system's potential as a context-aware tool for sustainable tourism. The results obtained in situ not only confirm the applicability of the model, but also provide user-centered insights for refining the recommendation strategy and enhancing the educational components delivered through mobile interactions.

5. Discussion

This study proposes a context-aware recommendation system that combines deep neural networks and ontology-based semantic modeling to support sustainable tourism in

the Santurbán paramo. Unlike earlier models focused solely on collaborative or content-based filtering, this approach integrates GeoSPARQL reasoning, local user preferences, and contextual attributes—such as distance, cost, and popularity—into an inference pipeline that functions entirely offline. This architectural design is particularly relevant in protected areas with low connectivity, where real-time server communication is not viable and autonomous decision-making becomes essential.

The system's capacity to operate locally and combine geospatial filtering with personalized recommendations differentiates it from approaches such as that of Bahrani et al. [81], who applied ontology-enriched filtering but relied on real-time network interactions. In contrast, the proposed method emphasizes autonomy and privacy by storing all user interactions and recommendations on the device. Furthermore, while models like those of Mou et al. [82], employ LSTM-based architectures to analyze route patterns, the current implementation adopts a dense feedforward network trained on features derived from an RDF ontology, which ensures interpretability and modular integration with semantic rules.

The integration of geospatial data and conservation alerts adds a novel dimension to the recommendation process. While Chatterjee et al. [83], applied deep learning and ontologies in eCoaching, their scope did not include spatial reasoning or ecological sensitivity. In contrast, the present system incorporates virtual geofences that trigger conservation-oriented messages upon entry into predefined zones, encouraging responsible behavior such as remaining on marked paths and avoiding disturbance to endemic flora and fauna. These mechanisms were evaluated through the system's deployment in the Santurbán region and complement prior works such as Bahramian et al. [77], who focused on semantic propagation but did not explore real-time ecological alerts.

Crucially, the field evaluation along the Laguna Negra trail (Section 4.3) provided direct feedback from users under real operating conditions. Participants consistently rated the system's usefulness, satisfaction, and diversity of recommendations between 4.1 and 4.4 out of 5, offering specific suggestions for improvement, such as enhancing the biodiversity content and navigation features. These results validate the alignment between the system's predictive logic and actual user needs. They also support the inclusion of ecological content in future iterations, as users expressed interest in richer information regarding flora and fauna.

In addition to visual alerts, the app employs auditory notifications triggered by geofence detection. These sound-based messages are designed to play only once upon zone entry, preventing repetition and avoiding intrusive user experiences. This modality is particularly valuable in environments where visual attention is limited due to terrain complexity or lighting conditions. The combination of context-awareness, user preferences, and conservation messaging aligns with sustainability goals by encouraging behavior that reduces ecological impact while enhancing educational engagement [84–86].

Ultimately, this discussion underscores the significance of combining machine learning with structured domain knowledge to produce recommendations that are not only technically sound but also socially and environmentally aligned. The findings from both quantitative performance and qualitative user evaluation demonstrate the feasibility of deploying intelligent, privacy-preserving, and conservation-oriented technologies in fragile ecosystems.

6. Conclusions

The implementation of a context-aware recommendation system tailored to the Santurbán paramo demonstrates the feasibility of delivering tourism guidance that aligns with principles of environmental sustainability and local autonomy. By integrating structured semantic information from an RDF ontology with a deep neural network inference pipeline,

the system delivers recommendations that respond to user preferences while remaining operational in offline conditions.

The local deployment strategy—based on SQLite and TensorFlow Lite—ensures that personal data remains confined to the user's device, which is particularly relevant for protected areas where connectivity is intermittent and privacy concerns are heightened. Through this approach, the system supports autonomous functioning without reliance on external servers, enabling the continuous provision of personalized recommendations and ecological alerts.

Evaluation results confirm that the model accurately estimates point-of-interest relevance using normalized features, such as distance, popularity, and cost. The combination of predictive metrics (RMSE, MAE, R^2, and AUC) and user feedback collected during field trials indicates that the system maintains consistent performance and practical usability in real settings.

The geofencing component, integrated with GPS and semantic reasoning, adds a spatial layer of environmental messaging. Contextual alerts linked to ecological boundaries offer an educational function by informing users about conservation practices and the presence of endemic species. These messages are triggered once per zone entry and include auditory signals, ensuring non-intrusive but effective delivery.

In practical terms, the system can support decision-making for visitors engaging with sensitive natural areas. Its modular structure allows for future expansion, including updates to the ontology, integration of new data sources, and adaptation to other protected regions with similar conservation needs. The proposed architecture thus offers a foundation for developing intelligent systems that balance personalized tourism experiences with environmental stewardship.

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Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: Data are contained within the article.

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Conflicts of Interest: The authors declare no conflicts of interest.

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